Overview

Overview Alerts 11 Reproduction

Dataset statistics

Number of variables	21
Number of observations	7043
Missing cells	11
Missing cells (%)	< 0.1%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	4.5 MiB
Average record size in memory	663.6 B

Variable types

Text	1
Categorical	12
Boolean	5
Numeric	3

Variables

Select Columns

customerID

Text

Unique

Distinct	7043
Distinct (%)	100.0%
Missing	0
Missing (%)	0.0%
Memory size	455.9 KiB

More details

gender

Categorical

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	371.5 KiB

More details

SeniorCitizen

Categorical

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	344.2 KiB

More details

Partner

Boolean

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	7.0 KiB

More details

Dependents

Boolean

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	7.0 KiB

More details

tenure

Real number (ℝ)

Skewed

Distinct	73
Distinct (%)	1.0%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	32.371149

Minimum	0
Maximum	72
Zeros	11
Zeros (%)	0.2%
Negative	0
Negative (%)	0.0%
Memory size	55.2 KiB

More details

PhoneService

Boolean

Imbalance Unique

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	7.0 KiB

More details

MultipleLines

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	385.1 KiB

More details

InternetService

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	385.5 KiB

More details

OnlineSecurity

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	378.2 KiB

More details

OnlineBackup

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	378.8 KiB

More details

DeviceProtection

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	378.8 KiB

More details

TechSupport

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	378.2 KiB

More details

StreamingTV

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	378.8 KiB

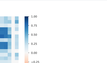
More details

StreamingMovies

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	378.8 KiB



More details

Contract

Categorical

Imbalance

Distinct	3
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	414.8 KiB

More details

PaperlessBilling

Boolean

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	7.0 KiB

More details

PaymentMethod

Categorical

Distinct	4
Distinct (%)	0.1%
Missing	0
Missing (%)	0.0%
Memory size	464.9 KiB

More details

MonthlyCharges

Real number (ℝ)

Imbalance

Distinct	1585
Distinct (%)	22.5%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	64.761692

Minimum	18.25
Maximum	118.75
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%
Memory size	55.2 KiB

More details

TotalCharges

Real number (ℝ)

Imbalance

Distinct	6531
Distinct (%)	92.7%
Missing	11
Missing (%)	0.2%
Infinite	0
Infinite (%)	0.0%
Mean	2283.3004

Minimum	18.8
Maximum	8684.8
Zeros	11
Zeros (%)	0.2%
Negative	0
Negative (%)	0.0%
Memory size	55.2 KiB

More details

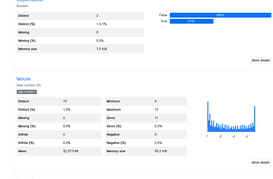
Churn

Boolean

Distinct	2
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	7.0 KiB

More details

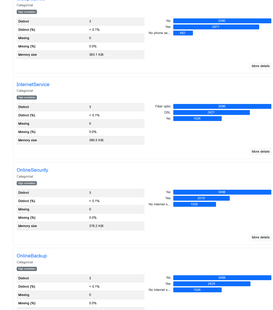
Interactions



Correlations

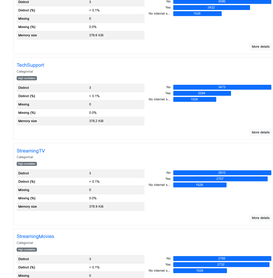
Auto

Heatmap Table



Missing values

Count Matrix



A simple visualization of nullity by column.

Sample

First rows Last rows

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProt
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No
5	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes
6	1452-KIOVK	Male	0	No	No	22	Yes	Yes	Fiber optic	No	Yes	No
7	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	Yes	No
8	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	Yes	Yes
9	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No